

When Does a Review Mean What It Says?

A DS 4002 Case Study by Austin Blackburn



Every day, millions of people log onto Letterboxd to rate and review the films they watch. A five-star rating signals a masterpiece and a half-star signals something far less generous; but what about the words in between? Reviews on Letterboxd are informal, expressive, and often deliberately satirical and ironic. A glowing paragraph can accompany a two-star rating, and harsh language may sit beneath four stars. This gap between language and rating demonstrates a fundamental challenge for any algorithm tasked with reading human opinion.

Pretrained sentiment models are widely used across industry to flag customer complaints, summarize feedback, and classify public opinion. But how well do they actually perform when the text is informal, sarcastic, emoji-laden, or written in a language they were not designed for?

You are a data scientist tasked with finding out. Using a dataset of Letterboxd reviews from the platform's Top 250 films, you will apply a pretrained HuggingFace transformer model and evaluate how well its sentiment predictions align with actual star ratings. Your analysis will go beyond a single accuracy number, investigating how review length, emoji usage, capitalization, and language each affect model performance.

Madanagopal, A. (2023). Letterboxd Top 250 Movies — Reviews Dataset. Kaggle.

<https://www.kaggle.com/datasets/alarchemn/letterbox-top250-short-reviews>

Barbieri, F., Camacho-Collados, J., Espinosa-Anke, L., & Neves, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. *Findings of EMNLP*.

<https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment-latest>